## 1. Summarising Notes

**Prompt**

You are an expert teacher.

Summarize the following notes into simple bullet points suitable for a beginner. Highlight the most important concepts and keep the summary under 200 words.

Notes:

Artificial Intelligence (AI) is the simulation of human intelligence by machines. AI systems can learn, reason, solve problems, and make decisions. Machine Learning is a subset of AI that enables systems to learn from data without explicit programming. Deep Learning is a subset of Machine Learning that uses neural networks to process large amounts of data.

**AI Response**

**Summary:**

- Artificial Intelligence (AI) enables machines to perform tasks that normally require human intelligence.
- AI can learn, reason, solve problems, and make decisions.
- Machine Learning (ML) is a branch of AI.
- ML allows computers to learn from data instead of being explicitly programmed.
- Deep Learning is a branch of Machine Learning.
- Deep Learning uses neural networks to analyze large amounts of data.
- AI → Machine Learning → Deep Learning represents a hierarchy of related technologies.

**Why This Prompt Worked**

This prompt clearly defined the role of the AI as an expert teacher and specified the format (bullet points), audience (beginner), and length limit. These instructions helped generate a concise and easy-to-understand summary.

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## 2. Generating a Study Plan

**Prompt**

Act as an academic mentor.

Create a 4-week study plan for a second-year Computer Science student preparing for GATE. The student can study 5 hours daily. Include topics, revision sessions, and practice questions.

Present the plan in a weekly format.

**AI Response**

**Week 1**

- Data Structures Basics
- Arrays and Linked Lists
- Stack and Queue
- Daily Practice: 5 Problems

**Week 2**

- Trees and Graphs
- Recursion
- Sorting Techniques
- Daily Practice: 5 Problems

**Week 3**

- Operating Systems Basics
- CPU Scheduling
- Memory Management
- Revision of Week 1 and 2

**Week 4**

- Computer Networks
- DBMS Basics
- Mock Tests
- Full Revision and PYQs

**Why This Prompt Worked**

The prompt clearly specified the student's background, study duration, target exam, and desired output format. This helped the AI generate a structured and relevant study plan.

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**3. Creating Interview Questions**

**Prompt**

Act as a professional resume writer.

Write a resume summary for a Computer Science student who knows Java, Python, Data Structures and Algorithms, Git, and basic AI concepts. The student is looking for internships.

Keep the summary professional and under 100 words.

**AI Response**

Computer Science student with strong foundations in Java, Python, Data Structures and Algorithms, and Git version control. Passionate about software development and problem-solving with hands-on experience in programming projects. Familiar with AI fundamentals and eager to apply technical skills in real-world environments. Seeking internship opportunities to contribute, learn, and grow as a software developer.

**Why This Prompt Worked**

The prompt provided the candidate's skills, target audience, purpose, and word limit. This enabled the AI to produce a concise and professional resume summary.

**5. Creating a Lesson Plan**

**Prompt**

Act as an experienced teacher.

Create a one-day lesson plan for teaching Arrays in Data Structures to beginners. Include learning objectives, topics, activities, and assessment methods.

**AI Response**

**Topic**

Introduction to Arrays

**Learning Objectives**

- Understand what an array is.
- Learn how arrays store data.
- Perform basic array operations.

**Topics Covered**

1. Definition of Arrays
2. Array Declaration
3. Array Traversal

4. Insertion and Deletion

**Activities**

- Classroom explanation
- Coding demonstration
- Practice exercises
- Quiz session

**Assessment**

- Multiple-choice quiz
- Coding exercise
- Question-and-answer session

**Why This Prompt Worked**

The prompt specified the audience, topic, and required sections. This helped the AI generate a complete lesson plan with clear educational structure.